

Chapter 122

VR Nodes

This chapter details the Virtual Reality (VR) nodes available in Fusion.

The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

Contents

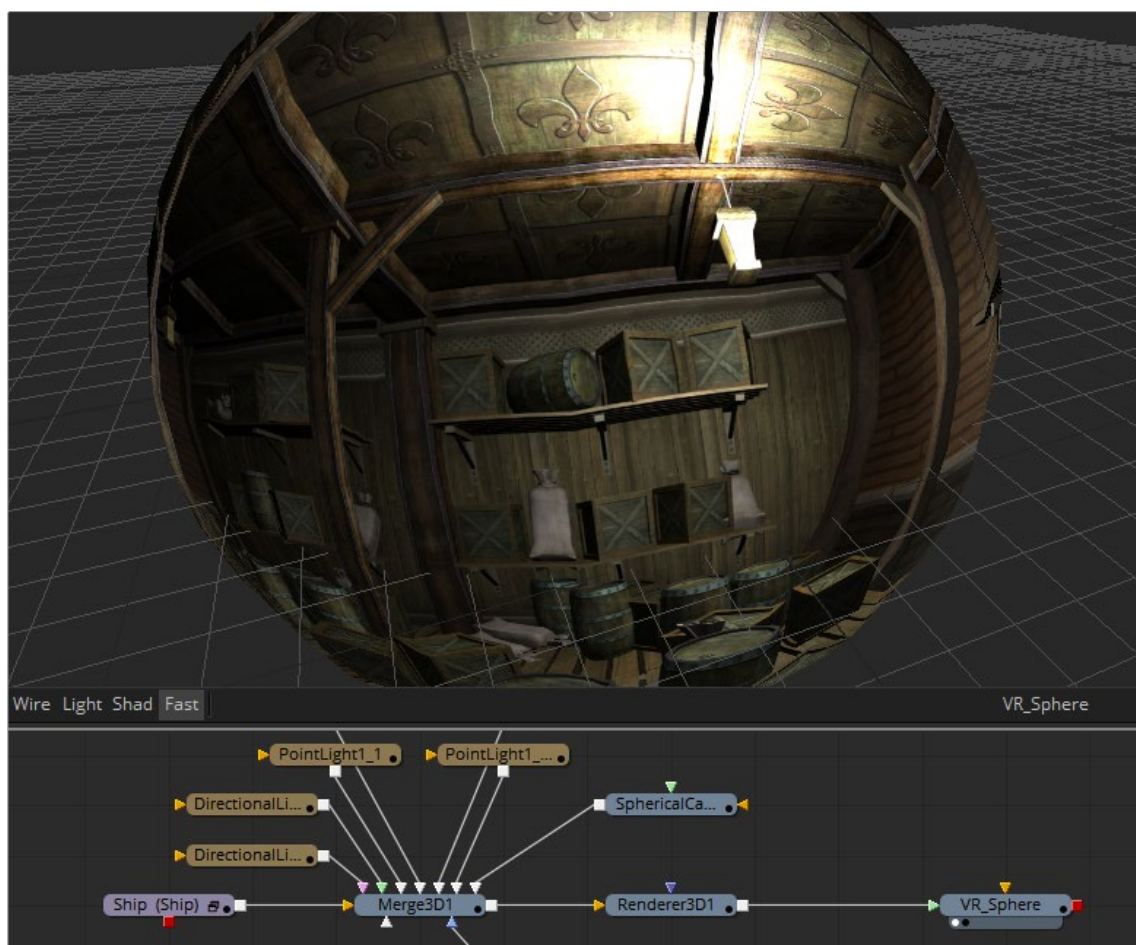
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VR Nodes

You can create and fix spherical (360°) video, often described as Virtual Reality, or VR, using Fusion's set of VR nodes. Dome productions, planetariums, and other special-venue theaters have utilized the flexibility of Fusion and its 3D system to produce and deliver special content for years.

The equirectangular (lat-long) format often used for 360° video is similar to how a globe is represented by a flat world map, with the poles at the top and bottom edges of the image and the forward viewpoint at the center.

TIP: You can create stereo VR using two stacked Lat Long images, one for each eye.



A VR Sphere displayed in the Node Editor and Fusion's 3D viewer

Fusion supports several common spherical image formats and can easily convert between them.

VCross and HCross: VCross and HCross are the six square faces of a cube laid out in a cross, vertically or horizontally, with the forward view in the center of the cross in a 3:4 or 4:3 image.

VStrip and HStrip: VStrip and HStrip are the six square faces of a cube laid vertically or horizontally in a line, ordered as Left, Right, Up, Down, Back, Front (+X, -X, +Y, -Y, +Z, -Z) in a 1:6 or 6:1 image.

LatLong: LatLong is a single 2:1 image in an equirectangular mapping.

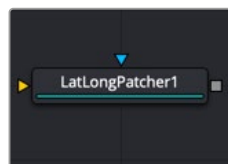
You can display both spherical video and live 3D scenes from the comp directly to headsets, including those from Oculus Rift and HTC Vive.

Fusion's "Fix it in post" tools for VR make it easy to perform several important tasks that are common in these types of productions.

Fusion VR 180 Support (Studio Version Only)

A number of Fusion's existing VR tools now support 180-degree angle of view. The 180-degree VR format allows you to produce immersive content and simplifies the production process for VR experiences by focusing on a forward-facing perspective.

Lat Long Patcher [LLP]



The Lat Long Patcher node

NOTE: The VR category and Lat Long node are available only in Fusion Studio and DaVinci Resolve Studio.

Lat Long Patcher Node Introduction

Equirectangular stitched images often need patches, paintwork, or other VFX applied. The Lat Long Patcher extracts and de-warps a section of a lat-long (equirectangular) image to be treated, and can warp and merge fixes back over the original. You can quickly pick a section of the spherical image to patch or paint, and then apply it back to the original image. Note that matching rotations are used in both Extract and Apply modes, allowing a node's operation to be easily reversed by a copy or instance with the same rotation settings.

Input

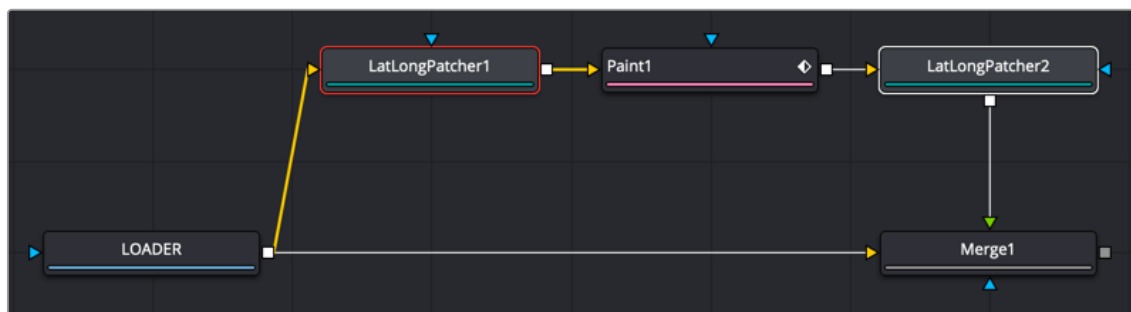
The Lat Long Patcher node includes two inputs. The orange input accepts a 2D image in an equirectangular format, where the X-axis represents 0–360 degrees longitude, and the Y-axis represents –90 to +90 degrees latitude. The effect mask input is provided, although rarely used on VR nodes.

Image Input: The orange image input accepts a equirectangular (lat-long) 2D RGBA image.

Effect Mask: The effect mask input is provided, although rarely used on VR nodes.

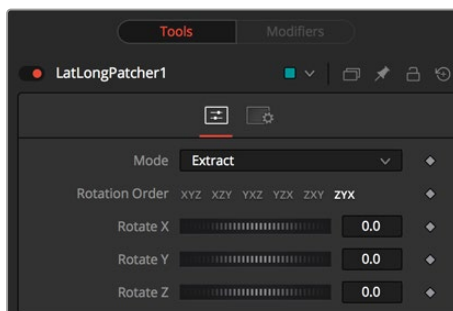
Basic Node Setup

The Loader node connects to the input on a Lat Long Patcher node. The output of a Lat Long Patcher node is set to Extract. It is then connected to whatever image-processing operation is required. A second Lat Long Patcher node set to Apply takes an input from the processed extraction and merges it over the top of the original source.



Two Lat Long Patchers used to repair a section

Inspector



The Lat Long Patcher Controls tab

Controls Tab

The Controls tab is used to extract and later reapply a section from an equirectangular image. Rotation controls allow you to select the exact portion you need to repair.

Mode

- **Extract:** Pulls a de-warped 90-degree square image from the equirectangular image.
- **Apply:** Warps and merges a 90-degree square image over the equirectangular image. Because the square image's alpha is used, this allows, for example, paint strokes or text drawn over a transparent black background to be applied to the original equirectangular image, avoiding any double-filtering from de-warping and re-warping the original.
- **Apply180:** Use to extract and apply a patch to a VR180 image. This gives control to apply traditional 2D effects, like paint, to a 180 distorted clip.

Rotation Order

These buttons choose the ordering of the rotations around each axis. For example, XYZ rotates first around the X axis (pitch/tilt), then around the Y axis (pan/yaw), and then around the Z axis (roll). Any of the six possible orderings can be chosen.

Rotation

These dials rotate the spherical image around each of the X, Y, and Z axes, offering independent control over pitch/tilt, pan/yaw, and roll, respectively.

Common Controls

Settings Tab

The Settings tab in the Inspector is duplicated in other VR nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Pano Map [PaM]



The Pano Map node

NOTE: The VR category and Pano Map node are available only in Fusion Studio and DaVinci Resolve Studio.

Pano Map Node Introduction

The Pano Map node converts images from one spherical layout to another, such as from a cube map to an equirectangular format. The node can also perform rotations of the spherical images when converting.

Input

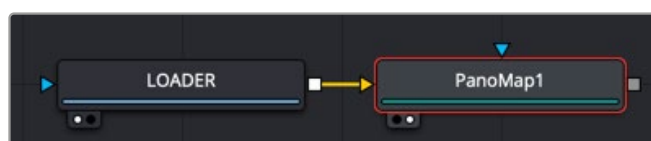
The Pano Map node includes two inputs. The orange input accepts a 2D image in an equirectangular, cube map or other spherical formats. The effect mask input is provided, although rarely used on VR nodes.

Image Input: The orange Image input accepts a spherical formatted 2D RGBA image that gets converted to another spherical format.

Effect Mask: The effect mask input is provided, although rarely used on VR nodes.

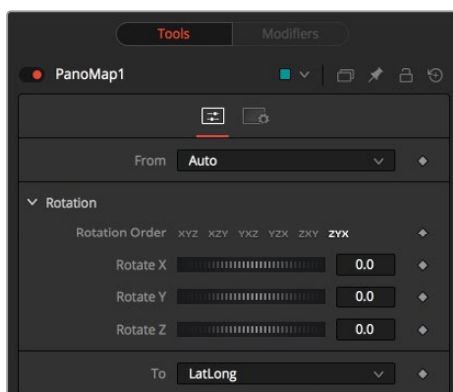
Basic Node Setup

In the example below, a Loader node containing a Lat Long image connects to the input on a Pano Map node. The Pano Map node is used to convert the image to an H-Cross format. It is then connected to whatever image-processing operation is required.



A Pano Map used to convert from one VR format to another

Inspector



The Pano Map Controls tab

Controls Tab

The Controls tab is used to determine the format of the input image and the desired output format.

From/To

- **Auto:** Auto detects the incoming image layout from the metadata and image frame aspect.
- **VCross and HCross:** VCross and HCross are the six square faces of a cube laid out in a cross, vertically or horizontally, with the forward view in the center of the cross in a 3:4 or 4:3 image.
- **VStrip and HStrip:** VStrip and HStrip are the six square faces of a cube laid vertically or horizontally in a line, ordered as Left, Right, Up, Down, Back, Front (+X, -X, +Y, -Y, +Z, -Z) in a 1:6 or 6:1 image.
- **LatLong:** LatLong is a single 2:1 image in equirectangular mapping.
- **VR 180** is a 1:1 180 degree stereoscopic image.

Rotation Order

These buttons choose the ordering of the rotations around each axis. For example, XYZ rotates first around the X axis (pitch/tilt), then around the Y axis (pan/yaw), and then around the Z axis (roll). Any of the six possible orderings can be chosen.

Rotation

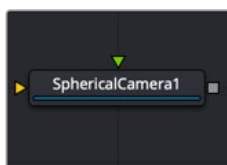
These dials rotate the spherical image around each of the X, Y, and Z axes, offering independent control over pitch/tilt, pan/yaw, and roll, respectively.

Common Controls

Settings Tab

The Settings tab in the Inspector is duplicated in other VR nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Spherical Camera [3SC]



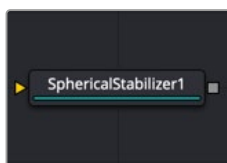
The Spherical Camera node

Spherical Camera Node Introduction

The Spherical Camera is not located in the VR category of the Effects Library but in the 3D category. However, it is commonly used in creating and fixing VR content, so it is referenced here. The Spherical Camera allows the 3D Renderer node to output an image covering all viewing angles, laid out in several different formats. This image may be used, for example, as a skybox texture or reflection map or viewed in a VR headset. The Image Width setting in the 3D Renderer sets the size of each square cube face so that the resulting image may be a multiple of this size horizontally and vertically.

For more detail on the Spherical Camera node, see *Chapter 89, “3D Nodes,”* in the DaVinci Resolve Reference Manual or Chapter 29, in the Fusion Reference Manual.

Spherical Stabilizer



The Spherical Stabilizer node

NOTE: The VR category and Spherical Stabilizer node are available only in Fusion Studio and DaVinci Resolve Studio.

Spherical Stabilizer Node Introduction

VR live action often uses handheld cameras, causing shaky footage to be a common problem. The Spherical Stabilizer node automatically identifies and tracks visible features in the footage, and then analyzes their movement to identify pan, tilt, and roll rotations. After tracking, it is then possible to smooth out or stabilize the rotation of the footage.

Inputs

The Spherical Stabilizer node has a single orange input.

Image: This orange image input node requires an image in a spherical layout, which can be any of Lat Long (2:1 equirectangular), VR180, Horizontal/Vertical Cross, or Horizontal/Vertical Strip.

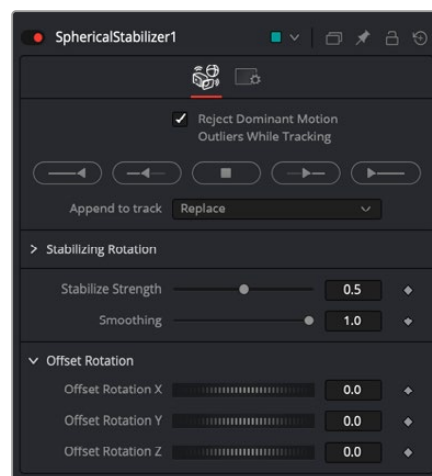
Basic Node Setup

In the example below, a 2:1 Lat Long image is connected to the input of the Spherical Stabilizer node. Once the image is stabilized, the output of the Spherical Stabilizer node is a steadied clip.



Spherical Stabilizer set up to steady a 2:1 Lat Long clip

Inspector



The Spherical Stabilizer Controls tab

Controls Tab

The Controls tab contains parameters to initiate the tracking and modify the results for stabilization or smoothing.

Reject Dominant Motion Outliers While Tracking

With this control activated (the default setting), features that move contrary to the majority of other features are ignored. This helps ignore the movement of subjects in the shot, preferring stable and consistent markers from the surrounding environment.

Track Controls

These buttons initiate tracking and analysis of the shot. Be aware that the reference frame used for stabilization is set to the first frame tracked.

- Track Backward from End Frame starts tracking backward from the end of the current render range.
- Track Backward from Current Time starts tracking backward from the current frame.
- Stop ceases tracking, preserving all results so far.
- Track Forward from Current Time starts tracking forward from the start of the current render range.
- Track Forward from Start Frame starts tracking forward from the current time.

Append to Track

- Replace causes the Track Controls to discard any previous tracking results and replace them with the newly-created track.
- Append adds the new tracking results to any earlier tracks.

Stabilization Strength

This control varies the amount of smoothing or stabilization applied, from 0.0 (no change) to 1.0 (maximum).

Smoothing

The Spherical Stabilizer node can eliminate all rotation from a shot, fixing the forward viewpoint (Still mode, 0.0) or gently smooth out any panning, rolling, or tilting to increase viewer comfort (Smooth mode, 1.0). This slider allows either option or anything in between.

Offset Rotation

Often a shot is not entirely level and needs the horizon to be realigned, or perhaps a desired pan should be reintroduced after fully stabilizing the shot. The Offset Rotation controls allow additional manual control of the Spherical Stabilizer's rotation of the footage, for pitch/tilt (X), pan/yaw (Y), and roll (Z), respectively. Rotation is always performed in the order X, Y, Z.

Common Controls

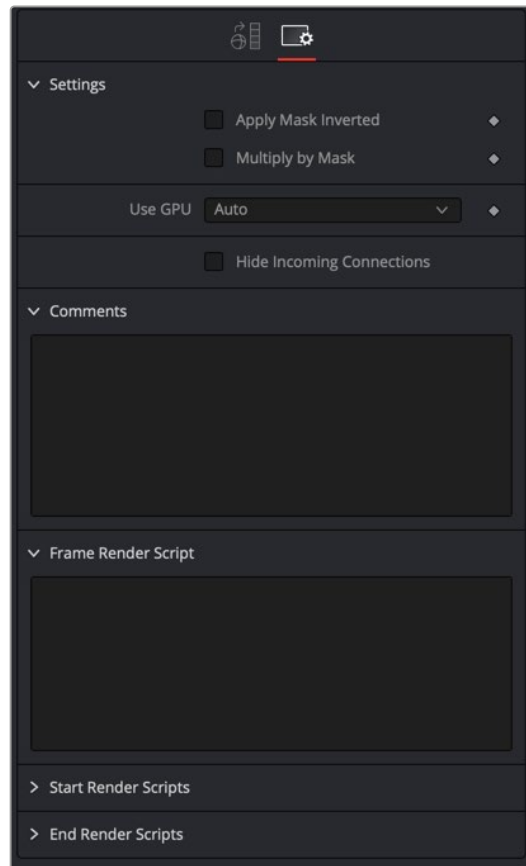
Settings Tab

The Settings tab in the Inspector is duplicated in other VR nodes. These common controls are described in detail in the following “The Common Controls” section.

The Common Controls

Nodes that handle VR operations share a number of identical controls in the Inspector. This section describes controls that are common among VR nodes.

Inspector



The VR Common controls

Settings Tab

The Settings tab in the Inspector can be found on every tool in the VR category. The controls are consistent and work the same way for each tool.

Apply Mask Inverted

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

Use GPU

The Use GPU menu has three settings. Setting the menu to Disable turns off hardware-accelerated rendering using the graphics card in your computer. Auto uses a capable GPU if one is available and falls back to software rendering when a capable GPU is not available.

Hide Incoming Connections

Enabling this checkbox can hide connection lines from incoming nodes, making a node tree appear cleaner and easier to read. When enabled, empty fields for each input on a node are displayed in the Inspector. Dragging a connected node from the node tree into the empty field hides that incoming connection line as long as the node is not selected in the node tree. When the node is selected in the node tree, the line reappears.

Comments

The Comments field is used to add notes to a tool. Click in the empty field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.